



Predicting early and complete drug release from long-acting injectables using explainable machine learning

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ABSTRACT

Polymer-based long-acting injectables (LAIs) have transformed the treatment of chronic diseases by enabling controlled drug delivery, thus reducing dosing frequency and extending therapeutic duration. Achieving controlled drug release from LAIs requires extensive optimization of the complex underlying physicochemical properties. Machine learning (ML) can accelerate LAI development by modeling the complex relationships between LAI properties and drug release. However, recent ML studies have provided limited information on key properties that modulate drug release, as existing approaches rely on time as a primary input feature, obscuring the independent contributions of material characteristics to release dynamics. This paper presents a novel data transformation and explainable ML approach to synthesize actionable information from 321 LAI formulations by predicting early drug release at 24, 48, and 72 h, classifying release profile types, and predicting complete release profiles. These three experiments investigate the influence of LAI material characteristics in early and complete drug release profiles. A moderate correlation (0.37) is observed between the true and predicted drug release at 72 h, while an F1-score of 0.72 is obtained in classifying the types of drug release profiles. For the first time, we demonstrate that time-independent ML frameworks achieve equivalent performance to time-dependent approaches in predicting complete drug release profiles, including complex delayed biphasic and triphasic curves. Shapley additive explanations reveal the relative influence of material characteristics during early time points, between drug release profile classes and for complete release, which fill several gaps in previous *in-vitro* and ML-based studies. The novel approach and findings can provide a quantitative strategy and recommendations for scientists to optimize the drug-release dynamics of LAI. The source code for the model implementation is publicly available in¹.

1. Introduction

Traditional research and development (R&D) efforts in industry and manufacturing are often impeded by high costs, low success rates, and prolonged research cycles (Xing et al., 2020). Compared to traditional methodologies carried out in laboratories, R&D based on artificial intelligence (AI) & D can better optimize product design and support accelerated material selection and function tests while reducing time and cost (Xing et al., 2020). In materials science, machine learning (ML), a subset of AI tools, can accelerate innovation by predicting the behavior and characteristics of materials based on their composition (Adetunla et al., 2024). Therefore, the characteristics of the target material can be rapidly prototyped using knowledge from limited laboratory data and advanced ML (Adetunla et al., 2024). A remarkable area of

materials science has led to next-generation drug delivery technologies (El-Tanani et al., 2025). Among these, long-acting polymer-based drug carriers improve therapeutic outcomes by reducing off-target toxicity while increasing the bioavailability of small molecules and biological drugs (Sultana et al., 2022). In this context, long-acting polymer-based injectable therapeutics require clinical approval for administration to treat chronic conditions such as neurological disorders, diabetes, and cancer (O'Brien et al., 2021; Witika et al., 2025). Before clinical approval, the development of novel long-acting polymeric drug carriers involves complex optimization of numerous formulation parameters (Park et al., 2019; Zulbeari et al., 2026), including particle size and drug encapsulation efficiency. The amount of drug released over time, known as the drug release profile, is a critical property of drug carriers. An optimized drug release profile is designed to maintain

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¹ https://github.com/mdsamad001/Drug_Release_Dynamics_Prediction/

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drug concentrations within a therapeutic window, the range between therapeutic and toxicity thresholds (Yoo and Won, 2020). For example, a burst release of a high concentration of drug (Park et al., 2019) can exceed toxicity thresholds, resulting in unwanted side effects (Yoo and Won, 2020). However, the phenomenon of burst release is difficult to study (Park et al., 2019) because it is usually revealed only by *in vitro* characterization without a regulatory definition. Furthermore, drug release profiling is the longest part of polymeric drug carrier characterization, which can take weeks to months. The variability in drug release duration, inconsistent sampling time, and heterogeneity in drug release requirements make it challenging to analyze the contributions of material characteristics to drug release profiles. Recent ML studies have used traditional methods and revealed a strong time-dependence in release profiles, thus overshadowing the relative contributions of material characteristics (Bannigan et al., 2023). We hypothesize that time-independent modeling of drug release curves can reveal material characteristic contributions that are obscured by time-dependent approaches. Drug release curves of LAI can vary in shape due to the influence of material characteristics. The time-independent curve of the release profile can reveal the contributions of LAI material characteristics that were previously obscured by time-dependent modeling. This paper presents a novel time-independent data transformation and a custom ML model to investigate the relationship between material characteristics and LAI drug-release profiles.

The organization of this paper is as follows. Section 2 provides a summary of current ML methods used to predict drug release. Section 3 outlines the proposed models and the experiments conducted in this study. Section 4 presents and compares the experimental results. Finally, Section 5 summarizes the key findings, discusses insights derived from the results, suggests possible directions for future research, and concludes the article.

2. Background

Long-acting injectables (LAI) improve the treatment of chronic diseases through sustained drug delivery, thereby reducing the frequency of drug administrations and the burden of treatment for patients (Baryakova et al., 2023). LAIs are developed to meet the unique therapeutic requirements of a target disease, including specific rates and concentrations of drug delivery. However, *in vitro* LAI optimization requires an iterative, costly, and time-intensive characterization of drug release rates because identifying key drivers of drug release remains a challenge (Park, 2025).

In this context, ML offers powerful computational solutions to modeling such multidimensional problems. Recent literature reviews have extensively examined the application of ML in biomaterial research (Meyer et al., 2022), including the study of polymer-based microparticles for drug delivery (Bao et al., 2025b) and drug release (Aghajanpour et al., 2025). Some applications of traditional ML predict microparticle characteristics such as particle size and drug loading (Alqarni and Huwaimel, 2025; Noorain et al., 2023; Rezvan-talab et al., 2024; Wang et al., 2024). Several groups have developed methods to predict partial (Husseini et al., 2025; Sun et al., 2025) and time-dependent (Bannigan et al., 2023; Catapano et al., 2025) drug release profiles. For example, Sun et al. (2025) applied traditional ML to predict cumulative drug release at discrete early time points up to 2 h from 97 PLGA nanoparticle formulations. Their data include only four input features: drug solubility, drug MW, particle size, and release matrix pH. Similar studies can be further advanced by designing data-specific custom ML models, increasing the number of formulations, incorporating a more diverse set of material characteristics, and extending the release profile prediction window to 72 h in most practical settings. A data-driven model that learns the relationship between material characteristics and drug release can be a valuable LAI development tool, substantially reducing the time and resources required for optimization.

This paper builds on recent successes (Deng et al., 2023; Bannigan et al., 2023; Sun et al., 2025; Catapano et al., 2025) of ML in modeling complex nonlinear relationships between biomaterial properties and drug-release dynamics. Among ML methods for predicting drug release profiles, decision tree-based methods are particularly successful (de Azevedo et al., 2017; Bannigan et al., 2023; Zawbaa et al., 2016; Catapano et al., 2025). For example, Bannigan et al. (2023) compared 11 traditional ML methods to predict drug release profiles, in which a light gradient-boosted tree model (GBDT) achieved the best performance. A key limitation of this and similar studies is their reliance on initial (Bannigan et al., 2023) or prior (Husseini et al., 2025) drug release information to forecast the subsequent drug release profile. The requirement of preceding experimental data in time and drug-specific criteria for release time are some impediments to developing robust models for discovering key material characteristics in this process. Our study differs from Bannigan et al. by removing time as an input feature for time-independent interpretation of material characteristics. Our proposed approach expands their ML prediction scope to include early fractional release and release profile stratification using one of the largest available data sets (Bao et al., 2025a) of 321 formulations. A methodological comparison between our and previous studies is provided in Appendix Table A1.

Few studies have explored the application of vanilla deep learning to model drug release profiles (Zawbaa et al., 2016; Salma et al., 2021; Husseini et al., 2025; Deng et al., 2023) without customizing for data-specific requirements. Collectively, these approaches have investigated the performance of various model architectures, including MLPs (68 formulations) (Zawbaa et al., 2016), ANNs (97 formulations) (Sun et al., 2025), DNNs (Husseini et al., 2025), CNNs, LSTMs, and GANs (Salma et al., 2021). Deng et al. (2023) have used complex deep learning models with 10 hidden layers and the ‘black-box’ ResNet architecture with 20 hidden layers to predict drug release for polymer-based microparticles (283 formulations). There are two major limitations of deep learning models. First, the sample size is often insufficient for deep learning to overcome overfitting. Second, the results of black-box deep models are challenging to interpret, but explainability is necessary to provide deeper insights into the prediction of drug release dynamics. A common approach in traditional and deep learning studies (Zawbaa et al., 2016; Salma et al., 2021; Husseini et al., 2025; Deng et al., 2023; Bannigan et al., 2023; Zhang et al., 2025; Catapano et al., 2025) is to use time values as input variables along with static material characteristics. Drug release certainly increases with time. However, the dependence of existing methods on time is so strong that they would not maintain predictive performance based solely on material characteristics. Recently, Catapano et al. (2025) predicted time-dependent drug release profiles using 113 PLGA microsphere formulations using the XGBoost model and its standard feature importance scores. However, standard feature importance scores from ML models do not explain the direction of change in an outcome with changes in input variables. Furthermore, their time-dependent approach precludes the study of contributions from material characteristics independent of time, which is a central methodological contribution of our work. Therefore, we argue that the strong influence of time in predicting the drug release profile, as presented in the SHAP feature analysis (Bannigan et al., 2023), overshadows the contributions of material characteristics to shaping the drug release profile.

This paper proposes a novel data transformation and a custom ML framework to predict the different curves of LAIs release profiles using material characteristics. The proposed framework addresses several key limitations of existing work, as summarized in Fig. 1 to make the following contributions.

- We build our models from multi-experiment and multi-source data (Bao et al., 2025a) to improve the reproducibility and generalizability of drug release prediction.

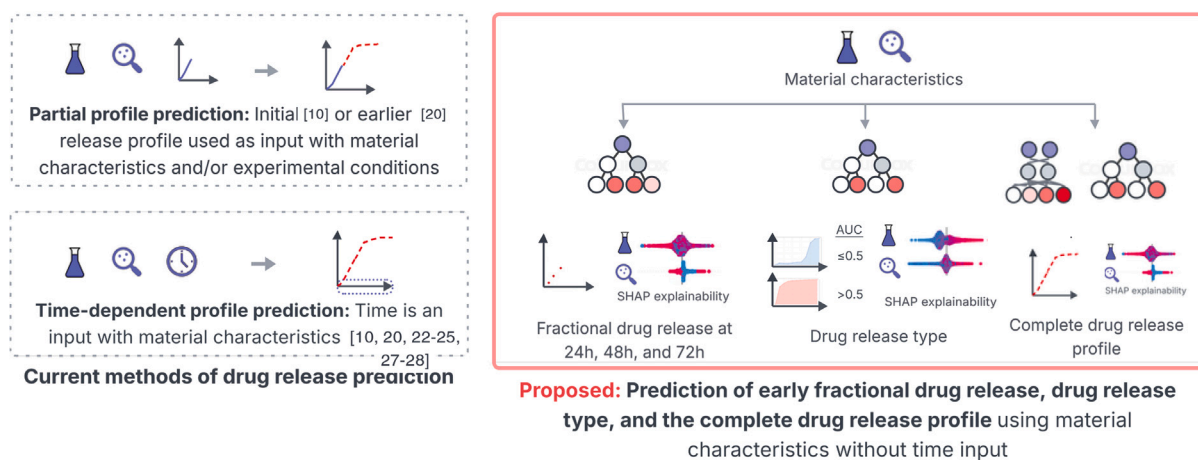


Fig. 1. Overview of standard approaches to predicting drug release and methodological contributions of this article.

- A time-independent predictive modeling approach is introduced in contrast to current time-dependent models that take time as an input feature.
- Fractional drug release is predicted at 24, 48, and 72-h time points to investigate the influence of material characteristics during the early release period when burst release possibly takes place.
- A novel area under the curve metric is proposed to characterize the time-independent curve of drug release. LAI drug release profiles are categorized into two distinct classes to investigate the effects of material characteristics in delaying drug release.
- The curve of complete drug release profile is predicted without preceding release information or requiring time as input.
- We have customized a recurrent neural network model to predict the complete curve of drug release profiles from the collective contributions of material characteristics.
- We present SHAP analyses to explain the overall and temporal influence of individual material characteristics for each predictive model. The relative influence of material characteristics on the drug-release profiles of polymer-based LAIs is presented and compared with current *in vitro* and ML studies.

3. Methods

This section provides a detailed description of the data set, data preparation, experiments, and evaluation of the models to predict drug release from LAI material characteristics.

3.1. Data set and data preparation

The data set of drug-loaded PLGA microparticles has recently been compiled from 113 publications using keywords such as PLGA microparticles, microspheres, and drug delivery (Bao et al., 2025a). The data selection criteria include (1) relevance to PLGA microparticles (MPs) for small molecule delivery, (2) completeness of the drug release profile from PLGA MPs spanning a minimum of 72 h and release exceeding 60%, and (3) PLGA MPs prepared using an emulsion-based method and reporting experimental details. The distribution of all formulation features used in model training is shown in Supplementary Figure S1.

The data set includes 321 drug release profiles of 88 drugs together with the corresponding formulation parameters and microparticle characteristics. The formulation frequency per drug is shown in Supplementary Figure S2. The distribution is right-skewed, with the majority of drugs having few formulations and a small number of drugs contributing disproportionately to the dataset, with an average of 3.6

formulations per drug. These formulation parameters and microparticle characteristics are listed in Table 1 and are used as input features to predict drug release. All features, except formulation methods, are numerical. The formulation method represents a nominal variable and is one-hot encoded prior to training the models.

The data set comprises drug release profiles covering a broad range of durations (minimum: 72 h, maximum: 238 days, mean: 30 ± 25 days) and measured at multiple time points. All release profiles reached over 60% of the maximum drug release. As the dataset was compiled from independent publications, dissolution methodology and temperature were not standardized across studies. However, solubility enhancer concentration in the release media was captured as a model feature to partially account for variability in release conditions. To ensure data uniformity in predictive modeling, drug release profiles are normalized and interpolated over time to produce distinct, comparable curves as follows. The duration of each drug release is scaled between 0 and 1 using Min–Max normalization, where each normalized time point (t_{norm}) is defined below.

$$t_{norm} = \frac{t - t_{min}}{t_{max} - t_{min}} \quad (1)$$

Here, t is the original time point, and t_{min} and t_{max} are the minimum and maximum time points of the drug release profile, respectively. Linear interpolation is applied to the drug release profiles to ensure uniformly sampled and fixed length input to the predictive models. Given two known release values y_0 and y_1 at time points t_0 and t_1 , respectively, a release value at a time point $t_0 < t_{int} < t_1$ can be interpolated as follows.

$$y_{int} = y_0 + \frac{t_{int} - t_0}{t_1 - t_0} \cdot (y_1 - y_0) \quad (2)$$

Thus, time interpolation maintains the rate of release at a given point to complete the curve of drug release profile.

3.2. Fractional drug release prediction

We perform a multifaceted experiment to investigate the contribution of LAI material characteristics to drug release. First, we interpolate fractional drug release at 24, 48, and 72 h to analyze early drug release. We choose up to 72 h because it is the minimum duration of drug release profiles in our data set. The first facet of machine learning for drug release is to predict fractional drug release at 24-h, 48-h, and 72-h time points. Regression models in machine learning, including linear regression (LR), random forest (RF), and extreme gradient boosting (XGB). Regression models will reveal the influence of individual material characteristics in the early stage of drug release.

Table 1

Summary of formulation parameters and microparticle characteristics in the data set.

Source: Adapted from Bao et al. (2025a).

Feature	Description
Formulation Method	Emulsion method used in microparticle preparation
Drug MW	Drug molecular weight
Drug TPSA	Drug topological polar surface area
Drug LogP	Drug log partition coefficient between aqueous and lipophilic phases
Polymer MW	Polymer molecular weight of the carrier
LA/GA	Polymer molar ratio of lactide to glycolide
Initial DPR	Initial drug-to-polymer weight ratio in microparticle formulation
Particle size	Diameter of microparticles
Encapsulation Efficiency	Percent of initial drug encapsulated in microparticles
Loading Capacity	Percent of drug-to-microparticle mass
Solubility Enhancer Conc.	Solubility enhancer concentration in <i>in-vitro</i> media

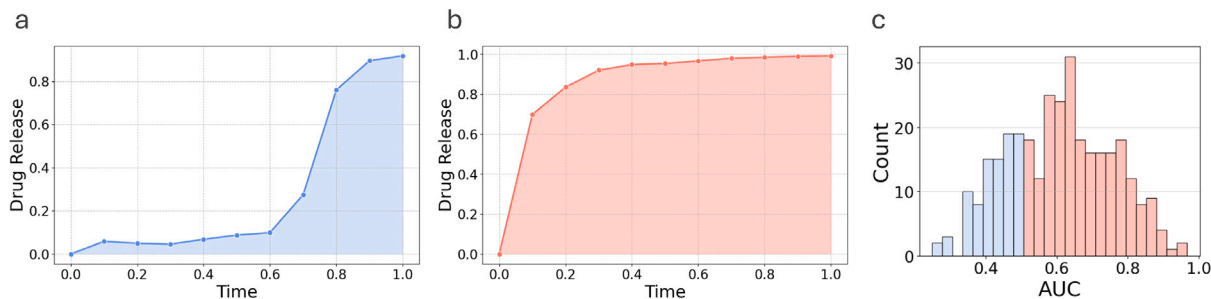


Fig. 2. Two distinct types of drug release profiles with (a) $AUC \leq 0.5$ and (b) $AUC > 0.5$. (c) The sample distribution based on the area under the curve (AUC) of release profiles. Blue for $AUC \leq 0.5$ and red for $AUC > 0.5$ (red) profile types.

3.3. Drug release classification

In the second experiment, we group drug release profiles in the data set based on the area under the curve (AUC). An AUC of 0.5 represents a monophasic or linear relationship between the normalized drug release profile and time. Monophasic, delayed biphasic and triphasic drug release profiles, as defined in Yoo and Won (2020), remain below the toxicity thresholds *in vivo*. Such profiles can be represented with $AUC \leq 0.5$, as shown in Fig. 2(a). On the other hand, burst biphasic profiles risk exceeding the toxicity thresholds *in vivo* (Yoo and Won, 2020). A burst biphasic (Yoo and Won, 2020) drug release profile is categorized by initial rapid release followed by slower release rates and can be represented by an $AUC > 0.5$, as shown in Fig. 2(b). The AUC threshold is a mathematical descriptor of the normalized drug release curve and does not imply a relationship between *in vitro* and *in vivo* pharmacokinetics or a Level A IVIVC assumption. The distribution of AUC values across the data set is shown in Supplementary Figure S3, indicating that the release profiles rarely fall below $AUC = 0.3$. The majority of formulations exceed $AUC = 0.5$, consistent with the predominant reporting of burst biphasic profiles in the literature. The sensitivity of classification performance to the AUC threshold is evaluated in Appendix Tabl A2, which indicates that an AUC of 0.5 best separates, and hence predicts, two distinct drug release profiles.

We conduct ML-based classification of two distinct drug release profile types: $AUC \leq 0.5$ versus $AUC > 0.5$, and investigate the contributions of material characteristics in differentiating profile types. Classifier versions of regression models, such as logistic regression (LR), RF, and XGB, are used for binary classification. The formulations in the training folds are adjusted to minimize the effect of class imbalance using random undersampling of the majority class. It is important to note that data structured in rows (formulations) and columns (material characteristics), as tabular data, are better predicted using traditional machine learning (such as decision tree-based models) than deep learning methods (Grinsztajn et al., 2022; Borisov et al., 2022; Abrar and Samad, 2022).

3.4. Time-independent prediction of complete drug release profile

In our third experiment, we predict the entire drug release profile using the static material characteristics presented in Table 1. No preceding or partial release data will be used to predict the drug release profile. Time and time-dependent variables are excluded to investigate the individual and collective contributions of material characteristics to a standardized representation of drug release dynamics. The goal of this experiment is to learn the relationship between the static material characteristics and the drug release profile.

For this experiment, we use two types of learning models: (1) baseline time-dependent models mostly used in the literature and (2) our proposed time-independent models. For the time-independent modeling, we develop a multi-output regression framework using multiple XGBoost models as XGB-multiregressor. Formally, given input data $X \in \mathbb{R}^{N \times d}$ with d static features of N formulations, a set of T independent XGBoost regressors is trained to jointly predict the uniformly sampled release profile $\hat{R} \in \mathbb{R}^{N \times T}$, where each regressor f_t learns the mapping $f_t: \mathbb{R}^d \rightarrow \mathbb{R}$ for time points $t \in \{1, \dots, T\}$. Each regressor is optimized independently to minimize the mean squared error at its respective time point. Unlike the time-dependent XGB model, which uses time as an input feature, XGB-multiregressor predicts the complete release curve solely from material characteristics, providing a time-independent tree-based learning solution alongside the recurrent neural network (RNN) framework.

The sequential release profile data are also predicted using a custom RNN model that captures the dynamic relationships between material characteristics and release over time without using time as an input feature. Therefore, RNN frameworks serve as time-independent solutions for learning drug-release profiles. RNNs are specifically designed to learn sequential data, such as a uniformly sampled release profile. However, basic RNNs suffer from vanishing or exploding gradients, which can cause long-term information to be lost. Long-short-term memory (LSTM) (Hochreiter and Schmidhuber, 1997) and gated with time units (GRUs) (Cho et al., 2014) address the limitations of basic RNNs. The conventional application of RNNs assumes time-series input data $X \in \mathbb{R}^{(N \times T \times d)}$ where N , d , and T represent the number of samples,

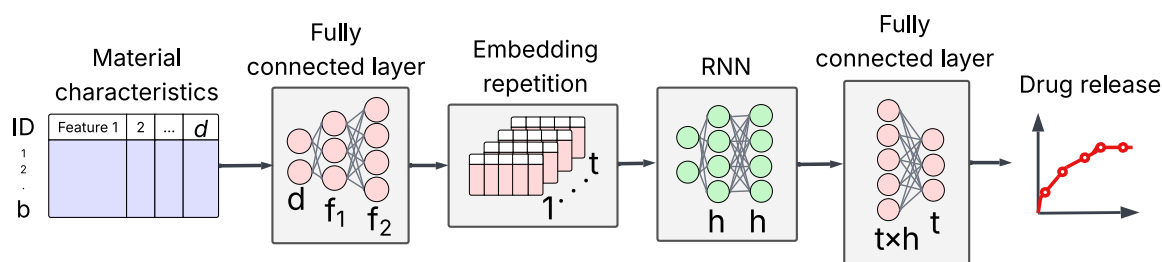


Fig. 3. Customized recurrent neural network (RNN) framework used for complete drug release profile prediction.

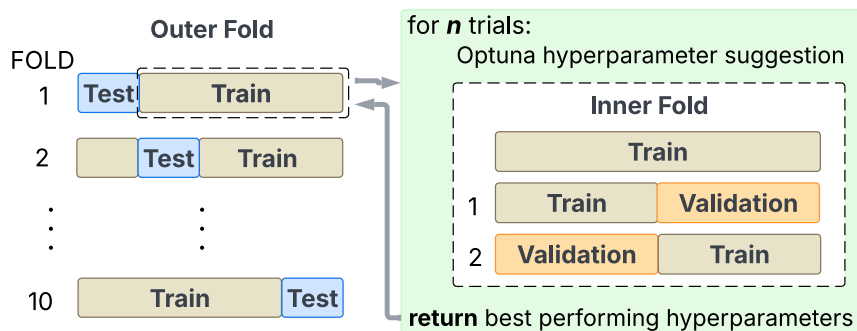


Fig. 4. Model validation and selection strategy using 10×2 nested cross-validation.

the number of features, and total points in a sequence, respectively. The output of RNN is $Y \in \mathbb{R}^{(N \times T \times h)}$ where h is the size of the hidden layer. However, in this study, static material characteristics are used to predict a uniformly sampled drug release profile. Therefore, a custom RNN-based framework is developed to learn the relationship between static characteristics and dynamic profiles, as shown in Fig. 3. Input data $X \in \mathbb{R}^{N \times d}$ with d static features of N samples are fed into a fully connected neural network (FC-NN) to obtain a new f dimensional feature space $Z \in \mathbb{R}^{(N \times f)}$. The f -dimensional space is repeated T times to create an input space $ZZ \in \mathbb{R}^{(N \times T \times f)}$ suitable for RNNs. The output of an RNN is $Y \in \mathbb{R}^{(N \times T \times h)}$, which is resized to $Y \in \mathbb{R}^{((N \times T) \times h)}$. The final layer of FC-NN projects Y onto the release profile $\hat{R} \in \mathbb{R}^{(N \times T \times 1)}$, which is reshaped to $\hat{R} \in \mathbb{R}^{(N \times T)}$. The RNN is optimized through backpropagation to minimize the sum of the mean squared error (MSE), as shown in Eq. (3).

$$MSE = \frac{1}{N} \frac{1}{T} \sum_{i=1}^N \sum_{t=1}^T (R_{i,t} - \hat{R}_{i,t})^2 \quad (3)$$

Here, the MSE is between the actual release magnitude (R) and the estimated magnitude $\hat{R}$ and averaged across T uniformly sampled points in a drug release profile.

3.5. Evaluation and explainability

A 10×2 nested cross-validation scheme is used to train and evaluate all machine learning and RNN models, as shown in Fig. 4. For ten outer folds, one test fold is left out every time for evaluation. Cross-validation folds are constructed such that all formulations of the same drug type, based on drug identity defined by molecular structure (chemical SMILES), are assigned to the same fold. Therefore, models are evaluated exclusively on unseen drugs, preventing leakage of drug-level physicochemical information across folds and providing a realistic estimate of model generalizability to novel drug candidates. Within the nine training folds, two inner folds are created to tune the model hyperparameters. Input features are standardized within the nine folds, which is applied to the test fold. The validation step identifies the best hyperparameter setting for model selection using the average scores across the two inner folds. We run the Optuna package (Akiba et al.,

2019) for 50 trials to find the best hyperparameter setting. The selected model is trained on nine-fold data and tested on the left-out fold.

The average scores across ten left-out folds are reported as model performance. In the fractional drug release prediction task at 24, 48, and 72 h, the performance of the regression models is evaluated using the root mean squared error (RMSE) between the actual (y) and predicted ($\hat{y}$) release as below.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

Where n represents the number of samples. We also report the Pearson correlation coefficient between the actual and predicted fractional drug release for the three time points. In the classification of drug release profile types, the model performance is reported using average classification accuracy, area under the receiver operating characteristic curve (AUROC), precision, recall, and F1 scores. In the third experiment, the predicted drug release profiles are evaluated using RMSE and correlation scores. The explainability of material characteristics for predicting fractional drug release at 24, 48, and 72 h, and for classifying drug release profile types, is provided using the Shapley additive explanations (SHAP) (Lundberg and Lee, 2017) method implemented in the shap Python library. SHAP scores for individual material characteristics quantitatively demonstrate their impact on drug release dynamics.

4. Results

All experiments are performed on Ubuntu 22.04, powered by an Intel(R) Xeon(R) W-2265 CPU (24 logical cores) running at 3.70 GHz, 64 GB of RAM, and a Quadro RTX A4000 GPU with 16 GB of video memory. The machine learning models are implemented and evaluated using the Sci-Kit learning package in Python. The RNN models are implemented using the PyTorch package in Python, which leverages parallel processing to automatically utilize multiple CPU cores.

4.1. Data and model preparation

Machine learning models have a set of hyperparameters that must be appropriately tuned for all prediction tasks. The hyperparameter

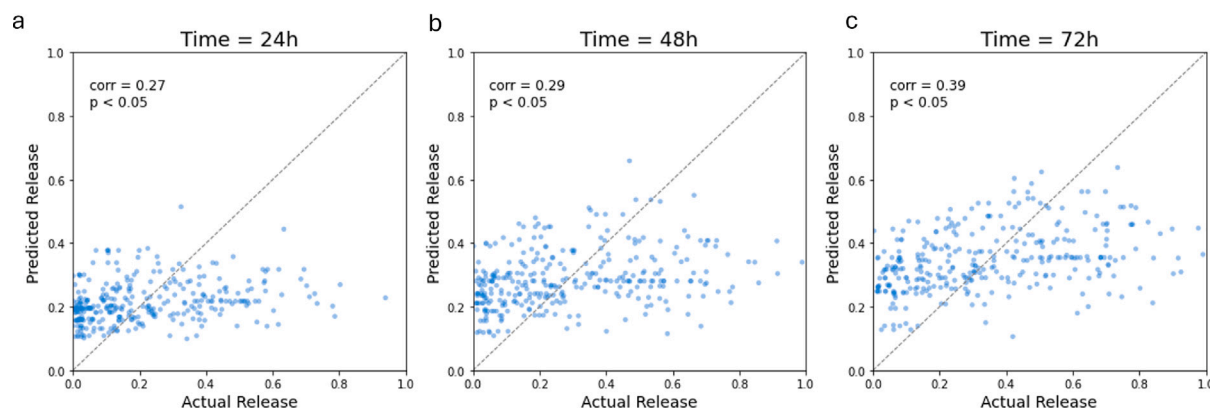


Fig. 5. Prediction of the drug release magnitude after (a) 24 h, (b) 48 h, and (c) 72 h following the onset of the drug release using the regression model (XGB).

Table 2

Performance scores of regression models in predicting fractional drug release at varying early release time points.

Model	24 h		48 h		72 h	
	RMSE	Correlation	RMSE	Correlation	RMSE	Correlation
LinR	0.19 (0.04)	0.21	0.22 (0.04)	0.21	0.24 (0.04)	0.24
RF	0.19 (0.04)	0.26	0.21 (0.04)	0.32	0.23 (0.04)	0.32
XGB	0.18 (0.04)	0.32	0.21 (0.04)	0.33	0.22 (0.04)	0.37

spaces used for machine learning and RNN models are as follows. The logistic regression model includes the inverse of regularization strength (C: 1e-4-1e4), penalty ('l1', 'l2'), solver ('liblinear'), maximum iterations (200-1000), class weight ('None', 'balanced'), and tolerance (1e-6-1e-3). Random forest hyperparameters include the number of estimators (50-300), maximum depth (3-20), minimum sample split (2-20), minimum samples leaf (1-10), maximum features (sqrt, log2), and bootstrap (True, False). The XGB hyperparameters include maximum depth (3-20), learning rate (0.01-0.3), number of estimators (50-300), column subsample by tree (0.5-1.0), gamma (0-5), regularization alpha (0-1), and regularization lambda (0-1). For the release profile prediction, RNN hyperparameters include hidden size (32, 64, 128, 256), number of layers (1, 2, 3), dropout (0.2, 0.3, 0.4), learning rate (1e-3, 1e-4), and batch size (16, 32, 64, 128). The number of epochs is determined empirically by training the RNN model up to 2500 epochs. The performance gain beyond 500 epochs of training appears to be negligible. Therefore, RNN models are trained for 250 epochs in the inner fold for model validation and 500 epochs in the outer fold for training, balancing model convergence against computational cost and to prevent overfitting.

4.2. Prediction of fractional drug release

The fractional drug release values at 24, 48, and 72 h are predicted from the formulation and microparticle characteristics using regression models to investigate predictors of early drug release. Table 2 presents the performance of the regression model.

Notably, linear regression (LinR) achieves comparable RMSE performance to non-linear tree-based models (RF, XGB) across all three time points, suggesting that material characteristics may share meaningful linear relationships with early drug release. However, tree-based models achieve higher correlation scores than LinR. RMSE scores are relatively lower at an earlier time point (24 h) than at a later time point (72 h). The correlation between predicted and actual fractional drug release at these time points is visualized in Fig. 5, showing scatter plots for (a) 24 h ($r = 0.32$, $p < 0.05$), (b) 48 h ($r = 0.33$, $p < 0.05$), and (c) 72 h ($r = 0.37$, $p < 0.05$).

Table 3

Average performance scores in classifying drug release profile types.

Model	Accuracy	AU-ROC	Precision	Recall	F1-score
LR	0.54 (0.11)	0.56 (0.19)	0.74 (0.17)	0.55 (0.18)	0.61 (0.16)
RF	0.66 (0.11)	0.57 (0.16)	0.77 (0.15)	0.72 (0.21)	0.72 (0.17)
XGB	0.65 (0.10)	0.52 (0.11)	0.77 (0.14)	0.70 (0.16)	0.72 (0.14)

Table 4

Average RMSE between the true and the predicted complete drug release profiles of two distinct types based on AUC threshold.

Model	AUC ≤ 0.5	AUC > 0.5	Overall
XGB-Time	0.205 (0.048)	0.162 (0.014)	0.177 (0.018)
XGB-No Time	0.332 (0.043)	0.314 (0.011)	0.320 (0.018)
XGB-Multiregressor - No Time	0.210 (0.04)	0.162 (0.014)	0.178 (0.014)
FC-NN-LSTM - No Time	0.190 (0.045)	0.180 (0.026)	0.189 (0.018)
FC-NN-GRU - No Time	0.188 (0.039)	0.176 (0.021)	0.183 (0.017)

4.3. Prediction of drug release types

Table 3 presents the performance of machine learning models in predicting drug release types. The data set consisted of 321 PLGA-based drug carrier formulations and the corresponding drug release profiles published by Bao et al. (2025b). Fig. 2(c) illustrates the class imbalance in drug release types under the selected AUC cutoff (AUC ≤ 0.5 : 84, AUC > 0.5 : 237). The class imbalance itself is not an artifact of the modeling approach but reflects the composition of the available literature, in which burst biphasic (AUC > 0.5) release profiles are disproportionately reported relative to sustained or delayed (AUC ≤ 0.5) release formulations (Yoo and Won, 2020) (Supplementary Figure S3). The random forest (RF) classifier model yields the best classification performance with an average F1-score of 0.72 (0.17), which is substantially better than the linear classifier model (LR). Low AU-ROC and accuracy scores across all models may be affected by class imbalance due to our strategy to maintain distinct drug types in training and test folds to avoid data leakage.

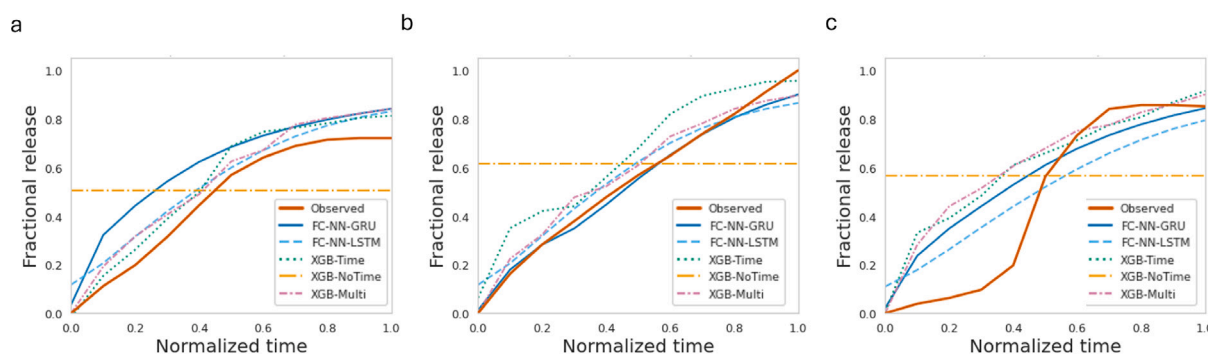


Fig. 6. Prediction of complete (a) burst biphasic (b) linear or sustained and (c) delayed triphasic drug release profiles using time-dependent (XGB-Time) and time-independent models (XGB-No Time, XGB-Multi, FC-NN-LSTM, FC-NN-GRU).

4.4. Complete drug release profile prediction

One of the contributions of our study is time-independent predictive modeling of complete drug release profiles from the static characteristics of materials by introducing a custom RNN-based learning framework (Fig. 3) and XGB-Multiregressor. Previous methods (Bannigan et al., 2023) have reported time as the primary predictor of drug release, which is certainly a time-dependent process. In contrast, our time-independent predictive models rely solely on material characteristics, without using time as an input variable. Table 4 presents the performance scores in the prediction of the complete drug release profile. The XGB-Time model, which takes material characteristics and time as inputs, yields the lowest error for release profiles with $AUC > 0.5$. However, both FC-NN-RNNs achieve better performance in predicting release profiles with $AUC \leq 0.5$. It is important to note that the XGB model fails without the time input, resulting in a flat drug-release profile and, ultimately, a high prediction error. In contrast, our time-independent models (FC-NN-GRU, FC-NN-LSTM, and XGB-Multiregressor) achieve competitive predictive performance without time as an input, a remarkable result given the absence of the primary predictor typically used in drug release prediction models. Notably, the time-independent XGB-Multiregressor matches the performance of XGB-Time across overall RMSE and AUC subgroups, confirming that time is not a necessary input for tree-based release profile prediction.

We further compare the model performance using pairwise Wilcoxon signed-rank tests on per-fold RMSE scores. No significant differences were observed between the FC-NN-GRU, FC-NN-LSTM, and XGB-Time models on overall RMSE (in Supplementary Figure S4 (a)) or adjusted R^2 (in Supplementary Figure S4 (b)) ($p > 0.05$ for all comparisons), indicating that the proposed time-independent models achieve predictive performance equivalent to the time-dependent XGB baseline.

Statistical tests are further performed within two subgroups of release profiles: (1) delayed or sustained profiles ($AUC < 0.5$) and (2) burst biphasic profiles ($AUC > 0.5$). The RNN-based architecture shows a specific advantage for delayed or sustained ($AUC < 0.5$) profiles, where temporal dependencies may be more complex. The time-independent XGB multi-regressor framework matches the time-dependent XGB performance, confirming that time is not a necessary input for competitive release profile prediction. However, the RNN-based architecture shows a specific advantage for delayed or sustained ($AUC \leq 0.5$) profiles, where temporal dependencies may be more complex. For example, FC-NN-GRU and FC-NN-LSTM achieved lower (better) RMSE than XGB-Multiregressor for delayed or sustained-release profiles, though this difference did not reach statistical significance, as shown in Supplementary Figure S4 (c). However, a comparison between time-independent models shows that our time-independent approaches (RNN, XGB-multiregressor) produce significantly better RMSE (lower) than their equivalent XGB-No time model ($p < 0.05$). For burst biphasic profiles ($AUC > 0.5$), Supplementary Figure S4 (d) shows

that our XGB-Multiregressor achieved significantly lower ($p < 0.05$) RMSE than both RNN models (FC-NN-GRU, FC-NN-LSTM). On the other hand, XGB-time significantly outperformed FC-NN-LSTM but showed no statistical difference relative to FC-NN-GRU. The superiority of tree-based models (e.g., XGB) on tabular data with static features, similar to ours, is well documented in the literature (Grinsztajn et al., 2022). In contrast, the RNN-based architecture facilitates learning complex temporal dynamics, such as those characteristic of delayed or sustained-release profiles.

Several representative release profiles for the two types are shown in Fig. 6. Fig. 6(a) shows that RNN-based models better capture the low AUC profiles ($AUC \leq 0.5$) where XGB-Time tends to overestimate. In contrast, RNN models tend to underestimate release profiles with high AUC scores ($AUC > 0.5$) compared to XGB-Time, as shown in Fig. 6(b). The statistical dependence between the model prediction error and the AUC scores of individual prediction models is presented in a box-and-whisker plot in Fig. 6(c). The figure shows that profiles with low AUC scores ($AUC < 0.5$) are better predicted by RNN-based models. Profiles with $AUC = 0.5$, representing monophasic of linear release, are predicted comparably by XGB-Time and the FC-RNN models. However, for high AUC scores, the prediction of the drug release profile becomes more time-dependent when the XGB-Time model performs the best. As shown in the same figure, without time information, the XGB model fails to achieve comparable predictive performance. In general, our results show that RNN-based models predict profiles with $AUC \leq 0.5$ robustly, suggesting that controlled release can be predicted from static material properties. Furthermore, the drug release curves associated with $AUC \leq 0.5$ such as delayed biphasic and triphasic curves, represent more complex drug release mechanisms than diffusion-based burst biphasic curves. FC-NN-RNNs capture these curves robustly from material characteristics alone. Prior drug release prediction studies have not explicitly evaluated model performance on such complex drug release profiles. The predominance of diffusion-driven burst biphasic curves with simpler more predictable kinetics may inflate the accuracy of time-dependent models. Evaluating models across diverse release types, as demonstrated here, provides a more rigorous assessment of predictive capability.

4.5. Explainable drug release dynamics

Contributions from material characteristics can further explain the robust prediction of drug-release dynamics. It can be argued that the time dependence of existing predictive models can overshadow the relative importance of material characteristics in drug release dynamics. Our proposed predictive models have achieved promising performance without time as input, suggesting their ability to learn drug-release dynamics from material characteristics.

The effects of individual material characteristics on fractional drug release are demonstrated by the SHAP beeswarm plots in Fig. 7. For all

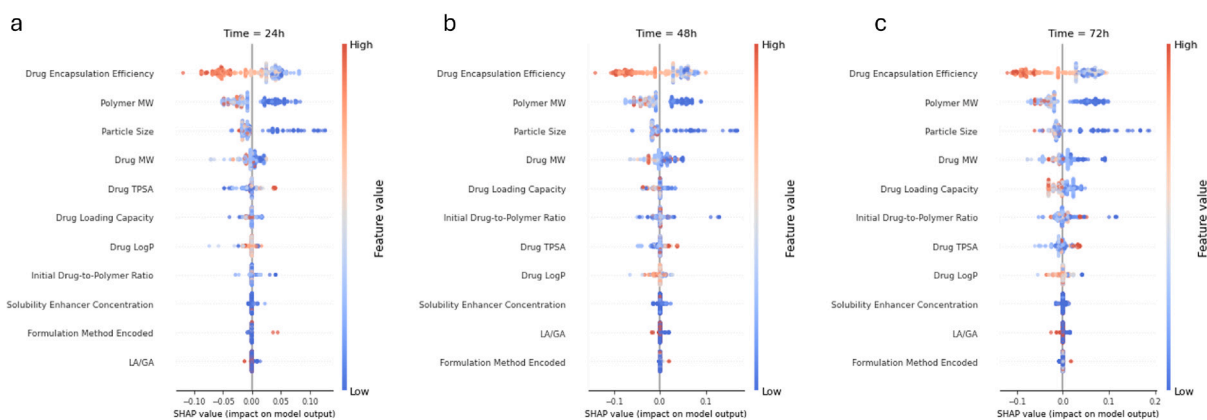


Fig. 7. Beeswarm plots for XGB regressor demonstrate SHAP feature impact on model prediction of fractional drug release at (a) 24, (b) 48, and (c) 72 h of drug release.

three time points (24-h, 48-h, 72-h), drug encapsulation efficiency appears as one of the most important characteristics during initial release. Higher drug encapsulation efficiency is associated with lower drug release. Next in the order, lower values of polymer MW and particle size are associated with higher fractional drug release. Higher values of drug TPSA and initial drug-to-polymer ratio are also indicative of higher drug release output. In terms of effect on drug release, drug MW, drug loading capacity, and drug lipophilicity rank next, with lower values of drug MW and loading capacity supporting lower fractional release. However, the effect of drug lipophilicity appears neutral. In general, LA/GA, solubility enhancer concentration, and formulation method demonstrate the lowest influence on drug release.

In classifying drug release profile types, the SHAP beeswarm plots in Figs. 8 a and b show the importance of individual material characteristics and explain their impacts on the prediction of the XGB and RF classifiers, respectively. Both classifiers demonstrate similar rank order of material characteristics in predicting the release profile types. Among the top ranked characteristics, higher values of drug LogP, loading capacity, and encapsulation efficiency are associated with the $AUC \leq 0.5$ release profile type, as reflected in lower output with $AUC \leq 0.5$ (release type = 0) than in release with $AUC > 0.5$ (release type = 1). In contrast, low Drug MW and Polymer MW values contribute to an increased likelihood of release with $AUC > 0.5$. The effect of particle size in the same direction is more prominent with the XGB classifier than with the RF classifier. A lower concentration of the solubility enhancer results in lower SHAP values, increasing the likelihood of $AUC < 0.5$ type release predicted by the RF classifier. In contrast, the effects of formulation methods, drug TPSA, and initial drug-polymer ratio are not prominent in terms of SHAP values.

To facilitate further quantitative insight, Fig. 9 presents a heat map of the mean absolute SHAP values showing the influence of individual material characteristics at different normalized time points of release using XGB-Multiregressor. Encapsulation efficiency is the dominant predictor throughout the full release profile, with its influence peaking in the early-to-mid phase of the release (Time: 0.1 to 0.5) before attenuating towards complete release. In the early-to-mid phase, polymer MW emerges as a strong secondary predictor alongside encapsulation efficiency. The influence of polymer MW here is consistent with its effect identified in Section 5.2.1, because polymers can control water influx and drug diffusion rates. Later, at mid-to-late time points, drug descriptors (drug MW, LogP) become the primary predictors after encapsulation efficiency, corroborating their influence on overall release profile type identified in Section 5.2.2. LA/GA ratio and formulation method remain low-importance throughout, likely overshadowed by the predictive strength of drug lipophilicity and molecular weight across the diverse formulations in this data set.

Collectively, these SHAP analyses demonstrate how material characteristics govern drug release. Encapsulation efficiency and polymer MW

are the primary drivers of early release, while drug lipophilicity and molecular weight gain increase in importance as the release mechanism transitions from diffusion- to erosion-based. This temporal shift in predictor importance has not been previously described in the ML literature on drug release, and offers a quantitative foundation for the recommended optimization outlined in Section 5.3.

5. Discussion

The development of LAI involves investigating an optimal combination of materials that yields the desired drug release dynamics, which is an iterative and costly process. To this end, this paper presents one of the first systematic investigations of the complex relationship between material characteristics and drug release dynamics, using data from multiple sources and drug formulations. The findings of the paper can be summarized as follows. First, the material characteristics of polymer-based microparticles contribute substantially to the prediction of early drug release, release profile types, and complete profiles. Previous studies have demonstrated this contribution on a limited scale. Second, despite limited samples, the correlation between true and predicted fractional drug release is between 0.21–0.37 within the early phase of drug release (24–72 h). Third, our methodological contributions demonstrate that material characteristics are predictive of drug release profile types, achieving up to a 0.72 F1-score in classifying drug release profiles. Fourth, one of the first efforts to predict an entire drug release profile demonstrates that an RNN-based framework is a better alternative to the time-dependent XGB model used in the literature in terms of accuracy and material contributions, especially in predicting more complex release profiles with $AUC < 0.5$, such as delayed biphasic and triphasic curves. Fifth, explainable ML methods reveal the strong and consistent effects of individual material characteristics on predicting early fractional drug release values and profile types.

5.1. Practical descriptors of drug release dynamics

The prediction of drug-release profiles of different types with varying release durations and inconsistent time points is a unique problem for data science. Despite widespread applications, traditional ML methods are not optimally designed to learn dynamic profiles from static variables. Current methods for predicting drug release profiles use an ad hoc time-based feature alongside material characteristics (Zawbaa et al., 2016; Salma et al., 2021; Hussein et al., 2025; Deng et al., 2023; Bannigan et al., 2023; Zhang et al., 2025) and demonstrate promising predictive performance. However, time-dependent predictive models obscure the importance of material characteristics that underpin drug release dynamics known from *in vitro* studies. Therefore, defining suitable target descriptors of drug release dynamics is a critical

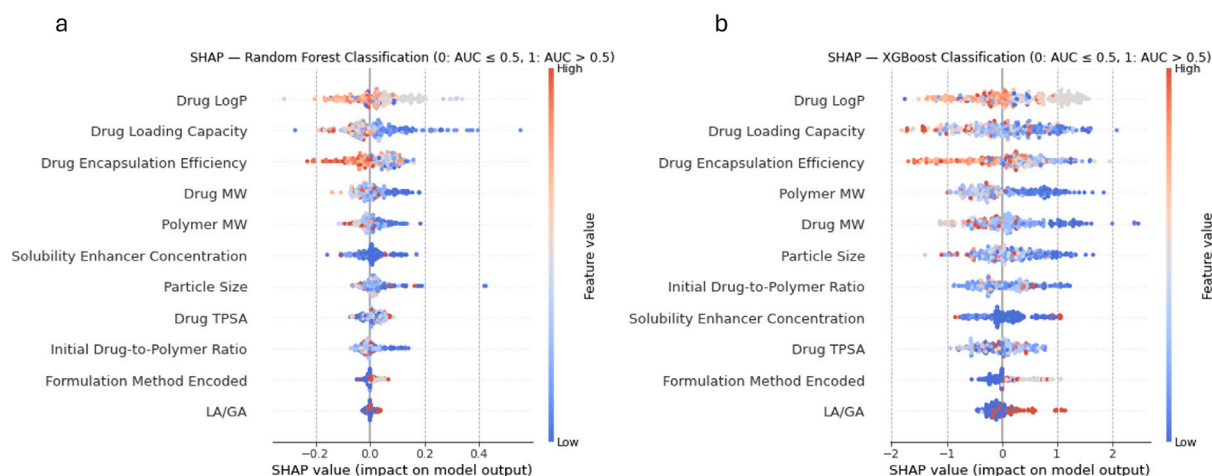


Fig. 8. Beeswarm plots for (a) RF and (b) XGB classifiers demonstrate SHAP impact on model prediction of drug release profile type (AUC $\leq$ 0.5: class 0, AUC $>$ 0.5: class 1).

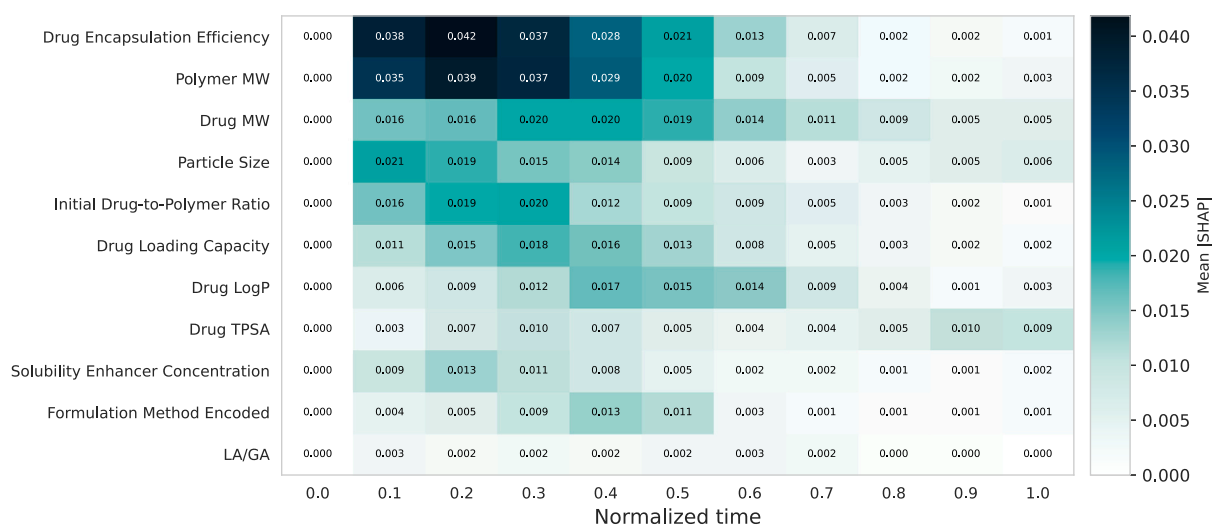


Fig. 9. Heat map showing the influence of individual material characteristics at varying time points (normalized) of drug release profiles. Darker colors indicate higher absolute mean SHAP values.

requirement for effective ML and a key contribution of this paper. Our time-independent data transformation and categorization of drug release profiles have allowed ML in heterogeneous formulations, decoupling predictive performance from time dependence and revealing the intrinsic contribution of material characteristics to drug release dynamics. In addition to predicting a global descriptor (AUC) of drug release profiles, we demonstrate the impact of material characteristics on early time points of drug release, which may facilitate LAI optimization to prevent burst release.

5.2. Effects of LAI material characteristics

Drug release is controlled by complex intermolecular interactions between the drug, the carrier, and the environment. In the literature, intermolecular interactions between the drug and the carrier have been reported to affect the distribution of the drug within the carrier (Park et al., 2019; Baumann et al., 2024; Yoo and Won, 2020), drug loading (Li and Yang, 2015), and the rate of drug release (Park et al., 2019). As a result of intermolecular interactions, hydrophobic drugs diffuse through the polymer matrix, and hydrophilic drugs are released primarily through diffusion through pores (Yoo and Won, 2020). Therefore, material characteristics drive intermolecular interactions, which,

in turn, determine drug release dynamics. However, the material characteristics of LAIs that drive intermolecular interactions are highly interdependent, which makes the prediction of drug release and the understanding of the relative contribution of individual characteristics *in-vitro* lab settings challenging.

Our proposed data transformation and explainable ML approaches address the limitations of previous studies to demonstrate the collective and relative effects of the individual material characteristics of polymer-based LAI on drug release. Most importantly, our method reveals a substantial predictive influence of material characteristics without being overshadowed by time or drug release after a certain time. Table 5 explains the rank and relative importance of material characteristics in different studies and contexts, which can help formulation scientists accelerate rational design and save valuable development time by focusing on the most important ones. Our results using explainable ML identify the leading predictors of early drug release between 24–72 h and predictors that differentiate between drug release profiles with low versus high AUC. In the following sections, we provide further insight into the effects of these predictors. Notably, while Bannigan et al. (2023) used SHAP for explainability, time-related variables accounted for more than 50% of predictive importance in their model, overshadowing the contributions of material

characteristics to drug release dynamics. Our time-independent framework resolves this limitation, revealing major contributions of material characteristics to drug release dynamics.

5.2.1. Effects on early drug release

Table 5 shows drug encapsulation efficiency, polymer MW, particle size, drug TPSA, and initial drug-to-polymer ratio are the leading predictors of early-stage drug release across all three time points. Higher values of drug encapsulation efficiency, polymer MW, and particle size are associated with lower early-stage release.

Encapsulation efficiency. Encapsulation efficiency refers to the percentage of initial drug encapsulated in the carrier which can be attributed to carrier fabrication influenced by intermolecular interactions (Li and Yang, 2015; Bickerton et al., 2012). In a recent ML study (Bannigan et al., 2023), the encapsulation efficiency was not reported for the prediction of drug release. However, our findings show that high values of drug encapsulation efficiency predict lower early release. Encapsulation efficiency can help describe drug-carrier stability, as high values of encapsulation efficiency indicate successful encapsulation and support low early-stage release (Luan et al., 2006), as supported by our findings.

Polymer MW. Our analysis also identifies high polymer molecular weight as a top predictor of low early-stage release. It has been reported in Feng et al. (2015) that high-MW polymers have longer polymer chains, leading to chain entanglements and reduced water influx, thus reducing polymer degradation and drug diffusion rates (Feng et al., 2015). Our results confirm that high values of polymer MW predict low early-stage release. Moreover, our results rank polymer MW as one of the leading predictors when considering numerous formulation variables.

Particle size. Furthermore, our findings identify smaller particle size as associated with increased early release. In the literature, one study reported that larger particle size was associated with higher drug release rates (Siepmann et al., 2004), while another reported results contrary to this observation (Klose et al., 2006). Our results show larger particles result in slower release in the early stages of release.

Drug TPSA. Drug descriptors such as topological polar surface area (TPSA) are intrinsically related to other molecular descriptors such as drug lipophilicity and molecular weight, and their influence is difficult to isolate or manipulate in *in-vitro* studies as a result. For example, drug analogs with different TPSA but identical MW or LogP values do not exist. Our results demonstrate high drug polarity (TPSA) is associated with increased early drug release prediction. Drugs with high TPSA are highly soluble in water and released primarily through diffusion, while prolonged release of non-polar drugs is supported through drug-polymer affinity and erosion-based release mechanisms. Importantly, these inferences are dependent on high encapsulation efficiency, which was a primary predictor across all models.

Initial drug-to-polymer ratio. The influence of initial drug-to-polymer ratio is also ranked among the top 5 for early drug release with the highest values being associated with increased fractional release. The initial drug-to-polymer ratio influences drug distribution throughout the polymer matrix. It is a formulation variable related to encapsulation efficiency and drug loading capacity. For example, with high drug-to-polymer ratio, particles can become overloaded and result in poor encapsulation and uncontrolled release.

5.2.2. Effects on drug release profile types

Table 5 shows that drug lipophilicity, loading capacity, encapsulation efficiency, and drug and polymer molecular weight are the primary predictive material characteristics of drug release profile types. Our findings suggest that there are some similarities and differences between predictors of early drug release and drug release profile types. In both early release and profile types, high values of encapsulation efficiency and polymer MW demonstrate strong influence in delaying the release.

Drug lipophilicity. Drug lipophilicity (LogP) refers to the ability of the drug molecule to dissolve in lipid versus aqueous environments. Drug lipophilicity also impacts intermolecular drug-carrier and drug-environment interactions to the extent that increasing lipophilicity can change controlled release from burst to delayed release, and lipophilic or fat-soluble drugs tend to release more slowly than water-soluble drugs (Baumann et al., 2024). However, the relative importance of drug lipophilicity among other material characteristics has remained unclear in the literature. For the first time in an ML study, our results demonstrate that high drug lipophilicity is a primary predictor of delayed drug release. In the data set, the mean of LogP 2.8 ± 2.8 with a median of 2.8 suggests that the data are at the border of slightly and moderately lipophilic drug LogP values. The LogP distribution is consistent with the physicochemical characteristics of small molecule drugs, which tend towards moderate lipophilicity to support permeability after drug delivery (Lipinski et al., 2001). We attribute the effects of increasing drug lipophilicity on delayed release to a transition in release mechanism by which the drug is released. For example, drugs with low lipophilicity values are released more easily by diffusion and thus associated with burst biphasic curves in the AUC > 0.5 class. On the other hand, drugs with high lipophilicity remain entrapped until the polymer erodes, represented in delayed biphasic or triphasic release profiles with an AUC ≤ 0.5 . The distribution of EE and LC across drug LogP values is shown in Supplementary Figure S5 (a) and (b), and the Spearman correlation (Supplementary Figure S6) between EE and drug LogP is 0.26 ($p < 0.001$), suggesting a moderate relationship between lipophilicity and encapsulation.

Drug content descriptors. Drug lipophilicity influences release mechanism and encapsulation as more lipophilic drugs tend to interact more favorably with hydrophobic PLGA matrix, supporting higher EE and LC values. A positive association between EE and LC in the dataset is shown in Supplementary Figure S5 (c), with a Spearman correlation of 0.49 ($p < 0.001$), (Supplementary Figure S6.) Our results identify high values of descriptors of drug-content as some of the top predictors of delayed release, namely drug encapsulation efficiency and loading capacity, both of which can be attributed to carrier fabrication influenced by intermolecular interactions (Li and Yang, 2015; Bickerton et al., 2012). Our findings show that high values of drug encapsulation efficiency and loading capacity significantly predict delayed release. It should be noted that, although increasing the loading capacity can aid in delayed release (Mao et al., 2007), carriers can become overloaded (Bickerton et al., 2012). Therefore, high loading capacity alone does not necessarily mitigate burst release (Ju Kim and Lee, 1992; Bickerton et al., 2012). Together, both drug content descriptors describe the amount of drug available and the stability of the encapsulation and are important predictors of delayed release.

Molecular weight. Our analysis also identifies high polymer and drug molecular weights as the top predictors of delayed drug release. Molecular weight (MW) refers to the weight of drug and polymer molecules, which is known to influence intermolecular interactions that regulate drug release. Our results confirm high values of polymer MW predict delayed release in early stages and in the overall drug release profile. Lower drug MW values can represent smaller drugs that are released through pores where larger drug molecules remain entrapped and take longer to diffuse. However, *in-vitro* studies have reported conflicting findings on the effect of drug MW on delayed release (Alhnan et al., 2010; Mao et al., 2007). This may be attributed to the limited scope of each study. Our results across diverse drug formulations suggest that high drug MW predicts delayed release. In an explainable ML study (Bannigan et al., 2023), polymer and drug MW are identified as the main predictors after time-related variables to predict the drug release profile. However, the magnitude of importance of these characteristics remains minimum because of the overpowering influence of time and early drug release information in their ML model.

Secondary predictors. The other material characteristics in our study that merit discussion include particle size, drug TPSA, lactide-to-glycolide ratio, (LA/GA) and formulation method. The influence of

Table 5

Comparison of factors important for predicting early drug release and drug release profile types using polymer-based LAIs. Delayed release indicates low early release and drug release profiles with $AUC \leq 0.5$. N.R.: not reported; N.S.: not significant. N.E.: neutral effect.

Factors	<i>in-vitro</i>	Baseline ML study (Bannigan et al., 2023)	Early drug release	Drug release profile types
Drug LogP	↑ Delayed release (Baumann et al., 2024)	N.S.	N.E.	↑ Delayed release
Encapsulation efficiency	↑ Delayed release (Luan et al., 2006)	N.R.	↑ Delayed release	↑ Delayed release
Polymer MW	↑ Delayed release (Feng et al., 2015)	↑ Delayed release	↑ Delayed release	↑ Delayed release
Drug TPSA	N.R.	↑ Delayed release	↓ Delayed release	N.E.
Loading capacity	↑ Delayed release (Mao et al., 2007) ↓ Delayed release (Ju Kim and Lee, 1992; Bickerton et al., 2012)	N.S.	↑ Delayed release	↑ Delayed release
Initial DPR	↓ Delayed release (Mishra et al., 2025)	N.E.	↓ Delayed release	N.E.
Drug MW	↓ Delayed release (Mao et al., 2007) ↑ Delayed release (Alhnan et al., 2010)	↑ Delayed release	↑ Delayed release	↑ Delayed release
Particle size	↓ Delayed release (Siepmann et al., 2004) ↑ Delayed release (Klose et al., 2006)	N.R.	↑ Delayed release	↑ Delayed release
Top 5 (rank)	Isolated effects.	Time, release at 24 h, Drug MW, Polymer MW, Drug TPSA	Encapsulation efficiency, Polymer MW, Particle size, Drug TPSA, Initial DPR	Drug Encapsulation efficiency, Polymer MW, Drug LogP, Drug MW
Relative importance	Relative contribution is unknown	Time-related variables account for >50% of SHAP values.	Top 3 material characteristics account for 30% of SHAP values	Top 3 material characteristics account for 30% of SHAP values

smaller particle size on uncontrolled drug release ranks higher for early drug release than for the overall profile. This can be attributed to a higher surface-area-to-volume ratio and a shorter diffusion path length, both of which facilitate initial drug release. While particle size remains a predictor of overall release, its influence is attenuated in cumulative release profiles. We speculate that the change in the behavior of the material characteristics is due to other competing factors that more decidedly predict drug release, according to Shapley additive explanations. In contrast to previous work on ML (Bannigan et al., 2023), our results do not find a significant predictive value for the topological surface area (TPSA) of a drug for the prediction of drug release type, despite its importance for early release prediction. The predictive importance of TPSA for early release, but not overall profile type, likely reflects a transition in the release mechanism from diffusion-based to erosion-controlled. Furthermore, a high lactide-to-glycolide ratio (LA/GA) is recommended to delay the release to reduce the likelihood of burst release (Yoo and Won, 2020). In contrast, the LA/GA ratio had a limited influence in our study compared to other factors on the prediction of delayed drug release. The limited influence of LA/GA can also be attributed to the considerable diversity of material characteristics in our study, particularly the wide spectrum of drug lipophilicity. Consequently, the magnitude of influence of the LA/GA ratio may be overshadowed by the predictive strength of other factors. Finally, our results suggest that the oil-in-water emulsion technique, typically used to encapsulate hydrophobic drugs, tends to predict delayed release.

5.3. Recommendations for materials scientists

The challenges in optimizing polymer-based long-acting injectables (LAI) arise from the complex and abundant underlying physicochemical

processes driving microparticle formation (Park et al., 2019) and drug release (Fredenberg et al., 2011). Studies on LAI formulations are growing and many variables have been shown to control drug release. However, due to constraints in laboratory experiments, each study is generally limited in scope to one or two material characteristics (Siepmann et al., 2004; Klose et al., 2006; Baumann et al., 2024). As a result, the combined contributions of material characteristics to drug release are not fully understood. Even when several material characteristics are considered for drug release experimentation (Mao et al., 2007), individual contribution of characteristics and complex relationship in between remain unknown. Therefore, researchers have recently explored ML in the prediction of drug release (Zawbaa et al., 2016; Salma et al., 2021; Hussein et al., 2025; Deng et al., 2023; Bannigan et al., 2023; Zhang et al., 2025) to investigate the contribution of individual characteristics to optimize drug release. This study advances knowledge in the field by addressing the limitations of previous studies to offer some recommendations on how to better control drug release with long-acting formulations.

We offer the following recommendations, which may serve as initial screening criteria rather than strict cutoffs. Researchers developing delayed release formulations should begin with materials selection, opting for hydrophobic candidates with higher (positive) values of drug lipophilicity (LogP: ~1–5 or higher) or low TPSA (< 60–90 Å²) and medium to high drug (> 300–400 Da or higher) and polymer (~50–100 kDa or higher) MW. LAI formulations should be optimized to maximize loading capacity (~10%–20%) and encapsulation efficiency (approx. > 70%) by screening the initial drug-to-polymer ratio prior to the characterization of the drug release profile. Notably, these general recommendations are not exclusive or exhaustive. For example, burst release can be reduced for a hydrophilic or low MW drug by increasing the encapsulation efficiency, polymer MW, or particle size.

Our findings recommend minimizing burst release by substituting or modifying drug candidates with higher lipophilicity and by selecting drug and polymer candidates with medium to high molecular weight. Secondly, optimizing both loading capacity and encapsulation efficiency by tuning the initial drug-to-polymer ratio is critical for controlling drug release. In addition, strategies for further fine-tuning the drug release profile should begin with encapsulation efficiency, drug and polymer molecular weight, followed by particle size.

5.4. Limitations

Our study is bounded by data and methodological constraints that should be considered. First, the early time points of 24, 48, and 72 h are chosen based on information on burst release scattered throughout the literature. Unfortunately, the criteria for burst release vary across drug types and therapeutic requirements. Therefore, there is no standard or regulatory criterion for burst release, which turns predictive modeling of burst release into a challenging task. Therefore, our data-driven results do not imply the prediction of burst release. Instead, our results should be interpreted as a general recommendation to prevent burst release and delay the drug release. Although time-independent representations of drug release profiles facilitate ML and the explainability of material characteristics, time information remains relevant to optimize precision drug delivery over time. Moreover, our method uses a dataset with limited samples from heterogeneous sources. The data do not include dissolution technique or temperature, which have also been shown to impact drug release kinetics.

6. Conclusion

This paper establishes a time-independent and explainable machine learning framework that identifies primary influencers of drug release from long-acting injectables. Our analysis reveals five key predictors of early drug release: encapsulation efficiency, polymer MW, particle size, drug TPSA, and initial drug-to-polymer ratio, and of drug release type: drug lipophilicity (LogP), loading capacity, encapsulation efficiency, and drug and polymer molecular weight. We demonstrate agreement with isolated effects reported in *in vitro* studies and present the relative importance of material characteristics driving early release and drug release types for the first time. Furthermore, we demonstrate that complete drug release profiles can be predicted with equivalent performance to time-dependent models without a time input, and SHAP analyses reveal evolving contributions of material characteristics at different phases of the drug release profile. For the first time, the complete prediction of drug release from polymer-based microparticles is achieved without a time-input feature. This work provides a practical tool for LAI development in which drug release profiles can be predicted according to the characteristics of the material.

CRediT authorship contribution statement

Karla N. Robles: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Manar D. Samad:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijpharm.2026.126922>.

Data availability

Data will be made available on request.

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